

INSTITUTE OF ACTUARIES OF INDIA

Subject SP4 - Pensions and Other Benefits

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**I. Defined Benefit (DB) Pension Scheme:**

- Benefits are predetermined based on a formula.
- Factors include final salary and years of service.
- Employer bears investment and longevity risks.
- Responsible for ensuring there are sufficient assets to meet benefits.

Defined Contribution (DC) Pension Scheme:

- Contributions are predetermined.
- Benefits are not predetermined.
- Member's benefits depend on investment performance.
- Member bears investment and longevity risks at the time of annuitization.

[4]

II. Assessment of Current Status:

- Conduct a thorough review of the DB scheme's financial status, including an actuarial valuation to determine any surplus or deficit.
- Evaluate the performance of the DC scheme and its ability to meet the retirement needs of employees.

Analysis of Impact on Employees:

- Consider the impact of restructuring on current and future employees' retirement benefits, ensuring that any changes are fair and equitable.
- Evaluate the potential benefits and drawbacks of merging the DB and DC schemes, converting the DB scheme to a DC scheme, or offering enhanced benefits in the DC scheme.

Regulatory and Legal Compliance:

- Ensure compliance with pension laws and regulations, including consultation requirements for any changes that affect employees' benefits.
- Seek legal advice to navigate complex regulatory issues and ensure that the restructuring is legally sound.

Communication Strategy:

- Develop a comprehensive communication plan to inform employees about the changes, including the reasons for restructuring and how it will impact their retirement benefits.
- Provide employees with the opportunity to ask questions and address any concerns they may have about the changes.

Options for Restructuring:

- Consider merging the DB and DC schemes to streamline administration and potentially reduce costs.
- Evaluate the feasibility of converting the DB scheme to a DC scheme to shift investment and longevity risks to employees.
- Explore the possibility of offering enhanced benefits in the DC scheme to provide employees with greater retirement security.

Implementation and Monitoring:

- Develop a timeline for implementing the restructuring plan, taking into account the need for regulatory approval and employee consultation.
- Monitor the implementation process to ensure that the restructuring is carried out effectively and that any issues are addressed promptly.

Review and Adjustment:

- Regularly review the restructured pension arrangements to ensure that they continue to meet the needs of employees and are aligned with ABC Corp's long-term financial and strategic goals.
- Be prepared to make adjustments to the restructuring plan as necessary to address changing market conditions and regulatory requirements.
- By following this comprehensive plan, ABC Corp can successfully restructure its DB and DC pension schemes to better align with its financial and strategic goals, while ensuring that employees' retirement benefits are protected.

[15]

[19 Marks]**Solution 2: Common Points for All Assumptions**

Document Rationale: Clearly document the rationale behind the chosen assumptions and any adjustments made, ensuring alignment with the plan's objectives and regulatory requirements.

Monitor Impact: Regularly monitor the impact of these assumptions on the plan's financial position and funding requirements, making adjustments as necessary to maintain alignment with economic conditions and the plan's funding goals.

Communicate Changes: Ensure effective communication of any assumption changes to all relevant stakeholders, including plan participants and regulators, to maintain transparency and trust.

I.

- Consider the current economic conditions, including low interest rates and market volatility.
- Review the yields of Government Bonds and/or high-quality corporate bonds matching the duration of the liabilities, as applicable.
- Assess the impact of a lower discount rate on the present value of liabilities.
- Consult with financial experts and review market trends for informed decision-making.
- Evaluate the impact of changes in the discount rate on the plan's funding requirements and financial position.
- Regularly review and update the discount rate assumption to reflect changing economic conditions.

[5]

II.

- Review the plan's current salary increase and benefit escalation assumptions.
- Consider the impact of higher inflation on future salary increases and benefit escalations.
- Use long-term inflation forecasts to adjust assumptions for future salary increases and benefit escalations.
- Account for the impact of higher inflation on the plan's liabilities.
- Adjust salary increase and benefit escalation assumptions to reflect the expected higher inflation.
- Regularly review and update assumptions for salary increases and benefit escalations to reflect changing economic conditions.

[5]

III.

- Analyze recent turnover data to assess the current turnover rate.
- Compare the current turnover rate with historical trends to identify any significant changes.
- Consider the impact of an aging workforce on retirement age assumptions.
- Review industry benchmarks and trends in turnover and retirement age.
- The assumptions may suitably be modified if there are heterogenous work groups

- Adjust turnover and retirement age assumptions to better reflect current and expected future trends.
- Account for factors influencing turnover and retirement age, such as economic conditions and company policies.
- Regularly review and update turnover and retirement age assumptions to reflect changing workforce demographics.

[5]

IV.

- Consider using a conservative approach to setting assumptions to minimize volatility.
- Use smoothed asset return assumptions to reduce the impact of short-term fluctuations in asset values.
- Adopt a stable and predictable discount rate assumption to minimize volatility in the present value of liabilities.
- Regularly review and update assumptions to reflect long-term trends rather than short-term fluctuations.
- Consider using asset-liability matching techniques to manage risk and volatility.

[5]

[20 Marks]**Solution 3:****I.** Here are six possible ways to distribute the surplus of a pension scheme:

- **Enhanced Benefits:** Provide increased pension benefits to members, such as higher monthly pension payments or additional lump sum payments.
- **Contribution Holidays:** Temporarily suspend or reduce employer and/or employee contributions to the pension scheme, allowing surplus funds to be used for other purposes.
- **One-Time Payments:** Provide one-time cash payments or bonuses to members, either as a lump sum or as a series of payments over time.
- **Benefit Improvements:** Make improvements to the pension scheme, such as offering additional investment options, improved death benefits, or enhanced early retirement provisions.
- **Surplus retained:** Use the surplus to as retained surplus in the scheme, ensuring the long-term financial health of the scheme.
- **Refund Contributions:** Refund a portion of members' contributions, either as a lump sum or as a reduction in future contributions, to reflect the surplus in the scheme.

[4]

II. **Fiduciary Duty:** The company and the trustees of the pension scheme have a fiduciary duty to act in the best interests of all scheme members. Any decision to provide enhanced benefits to certain groups must be justifiable in terms of this duty.

Trust Deed and Rules: The company should review the trust deed and rules of the pension scheme to ensure that it has the legal authority to make changes to benefits and to determine the criteria for eligibility for enhanced benefits.

Consultation Requirements: Depending on the nature of the enhancements, the company may be required to consult with affected members and/or employee representatives before making any changes.

Pensions Regulator Guidance: The company should consider any guidance issued by the Pensions Regulator regarding the use of surplus funds and the provision of enhanced benefits.

Financial Reporting Standards: The company should ensure that any enhancements are accounted for in accordance with applicable financial reporting standards, such as IAS 19 or FRS 102.

Legal Advice: The company should seek legal advice to ensure that any enhancements are implemented in compliance with all relevant legal and regulatory requirements.

Communication: The company should communicate clearly and effectively with affected members about the enhancements, including the criteria for eligibility and any potential impact on other members and how to handle conflicts.

Actuarial Aspects:

- Experience Analysis: Conduct an experience analysis to identify the sources of surplus, whether due to factors like favorable investment returns or lower-than-expected claim rates.
- Surplus Allocation by Source: Allocate the surplus according to its sources, ensuring that funds earmarked for specific sources (e.g., excess returns) are appropriately used.
- Materiality and Surplus Amount: Assess the materiality of the surplus amount to determine the feasibility of enhancements, ensuring that the allocation does not jeopardize long-term funding.
- Future Appropriations: Consider reserving a portion of the surplus for future appropriations to address potential future liabilities or adverse conditions.

By incorporating these legal, financial, and actuarial considerations, the company can make an informed, compliant, and financially sound decision on surplus usage.

[10]

III.

Benefits for the Company:

Reduces long-term pension liabilities, potentially improving the company's financial position. Provides a way to manage pension costs and risks, especially if the company is struggling financially.

Improves the predictability of future pension costs by eliminating future pension increases.

Risks for the Company:

May face legal or regulatory challenges if members feel pressured to accept the offer.

Could lead to reputational damage if members perceive the offer as unfair or coercive.

Future financial circumstances of the company could change, impacting its ability to meet ongoing pension obligations.

Benefits for the Scheme:

Reduces the scheme's liabilities, making it more secure and potentially improving its funding position.

Provides members with a choice and potentially a valuable financial opportunity.

Could help the scheme align its liabilities more closely with its assets, reducing funding volatility.

Risks for the Scheme:

Members who accept the offer lose the future pension increases they would have received, which could impact their retirement income.

The scheme may face backlash if members feel they were not adequately informed about the implications of the offer.

There could be administrative complexities in implementing the exchange exercise.

risk of anti selection – members in ill health may choose to exercise the option.

Benefits for Members:

Provides members with a one-time cash payment, which could be used for immediate financial needs or investments.

Gives members the flexibility to choose between a lump sum payment and future pension increases based on their individual circumstances.

Reduces the risk of future pension increases not being granted due to financial constraints.

Risks for Members:

Members who accept the offer lose the security of future pension increases, which could impact their standard of living in retirement.

The cash payment may not fully compensate for the value of the future pension increases lost.

Members may not fully understand the implications of the offer and could make a decision that is not in their long-term financial interest.

Possibility of splurging the lumpsum on non-essential things which would have otherwise been used for old age benefits

Legal and Regulatory Considerations:

The offer must comply with pension laws and regulations, including ensuring that members are not unfairly disadvantaged.

The company and trustees have a fiduciary duty to act in the best interests of the scheme members, which may be scrutinized in this context.

Communication and Engagement:

Clear and transparent communication is essential to ensure that members understand the offer and its implications.

Engaging with members and addressing their concerns can help mitigate risks and build trust in the decision-making process.

Financial Impact:

The one-time cash payment may have tax implications for members, which should be clearly communicated.

The financial impact of the offer on the company's balance sheet and cash flow should be carefully evaluated.

Long-term Sustainability:

The offer should be part of a broader strategy to ensure the long-term sustainability of the pension scheme.

Consideration should be given to how the offer aligns with the company's overall financial and strategic objectives.

Member Engagement and Support:

Providing members with access to independent financial advice can help them make informed decisions.

Offering support to members who may be vulnerable or uncertain about the offer can help ensure fairness and transparency.

Impact on Future Pension Provision:

The offer could impact the scheme's ability to provide future pension increases or other benefits, which should be considered in the decision-making process.

The long-term implications of the offer on the scheme's funding position and ability to meet future obligations should be evaluated.

Employee Relations:

The offer could impact employee morale and perception of the company's commitment to employee benefits.

Clear communication and engagement with employees can help manage expectations and minimize negative impacts on employee relations.

Trustee Responsibilities:

Trustees have a duty to act in the best interests of scheme members and should carefully consider the implications of the offer.

Trustees should seek independent advice if necessary to ensure that they are fulfilling their fiduciary duties.

Regulatory Scrutiny:

The offer may attract regulatory scrutiny, especially if there are concerns about the fairness or transparency of the process.

Compliance with regulatory requirements and guidelines is essential to avoid potential legal or reputational risks.

Impact on Scheme Funding and Investment Strategy:

The offer could impact the scheme's funding position and investment strategy, which should be reviewed in light of the proposed changes.

Trustees should consider the long-term implications for the scheme's funding and investment strategy when evaluating the offer.

Financial Advice and Guidance:

Providing members with access to financial advice and guidance can help them make informed decisions about the offer.

Ensuring that members understand the implications of the offer for their retirement income is essential.

Potential for Member Discontent:

Members who do not participate in the offer may feel disadvantaged compared to those who do, which could lead to discontent among members.

Clear communication and engagement with all members can help address concerns and minimize the risk of discontent.

Impact on Scheme Governance:

The offer could impact the governance of the scheme, including the role of trustees and the relationship between the company and the scheme.

Trustees should ensure that they have the necessary expertise and resources to evaluate the offer effectively.

Monitoring and Review:

The offer should be monitored and reviewed regularly to assess its impact on the scheme and its members.

Feedback from members should be sought to ensure that any concerns or issues are addressed promptly.

(20)

[34 Marks]

Solution 4:

I. The trustees should take the following steps:

When considering the strength of the sponsor's covenant the trustees can monitor the sponsor as follows:

- reviewing publicly available financial metrics such as the sponsor's published accounts
- taking into account any risk-based measure, for example the sponsor's credit rating if this is available. For example, there may be a risk-based premium to pay to a centrally run compensation fund, the size of the premium can be used as an indicator of strength.
- meet regularly with the finance director or board of the sponsor to discuss its financial position and any plans for the future
- imposing covenants on the sponsor for notifying the trustees of circumstances that could materially affect the security of members' benefits
- undertake regular reviews of the strength of the sponsor's covenant with reference to qualified professionals.

[5]

II. Valuation Adjustments:

Discount Rate Adjustment: Lower the discount rate to reflect the higher default risk of the sponsor. This adjustment results in a higher present value of liabilities, creating a more conservative valuation.

Incorporate Probability of Default: Use quantitative models to estimate the probability of sponsor default, impacting the discount rate and overall liability valuation.

Recovery Rates: Consider expected recovery rates in the event of sponsor insolvency to adjust the valuation accurately.

Funding Adjustments:

Value at Risk (VaR): Implement VaR, integrated with stochastic modelling of interest rate assumption, to quantify the potential shortfall in funding due to fluctuations in investment returns and interest rate movements. VaR helps assess the maximum potential loss in the scheme's funding level with a specified confidence interval over a set timeframe. For example, "We are 95% confident that the funding level will not fall below 80% over the next 5 years."

By simulating various interest rate paths, VaR can quantify how changes might impact funding levels and the sponsor covenant over time. This ensures that discount rates reflect both current market conditions and projected volatility.

Higher Contributions: Recommend increasing contribution rates from the sponsor to improve the funding level rapidly. This can be structured with contingent contributions that increase if the sponsor's financial position deteriorates further.

Contribution Ratchets: Include ratchets in the contribution schedule to ensure that contributions increase if the sponsor's financial condition improves.

Investment Strategy Adjustment: Shift the investment strategy towards lower-risk assets to reduce the volatility and better match the scheme's liabilities.

Bonds and Credit Default Swaps: Increase the allocation to high-quality bonds and consider investing in credit default swaps (CDS) to hedge against the risk of sponsor default.

Specific Steps for Trustees:

Regular Monitoring: Continuously monitor the financial health of the sponsor using publicly available financial metrics, credit ratings, and regular meetings with the sponsor's finance team.

Impose Covenants: Require the sponsor to notify trustees of any significant changes in its financial condition that could affect the scheme's security.

Engage Professionals: Seek advice from financial advisors and actuaries to reassess the scheme's funding and investment strategies regularly.

Independent Covenant Assessment: Hire specialists to perform independent assessments of the sponsor covenant and incorporate the findings into actuarial advice.

Contingency Planning: Develop and implement contingency plans to protect the scheme's benefits in the event of sponsor insolvency.

Securing Additional Guarantees: Obtain additional security, such as charges over fixed assets or setting up escrow accounts, to protect the scheme's interests.

Communication: Maintain transparent communication with scheme members about the sponsor's financial difficulties and the measures being taken to safeguard their benefits.

Risk Disclosure: Inform members about the potential risks to their benefits and the steps being taken to mitigate these risks.

By using techniques like Value at Risk (VaR) in funding advice and other quantitative methods, actuaries can provide more precise and realistic recommendations. Trustees can then take informed actions to manage the risks associated with a financially distressed sponsor and ensure the long-term sustainability of the pension scheme.

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III. Insurer Selection:

Financial Strength: Choose an insurer with a strong financial rating and a proven track record in managing pension liabilities.

Due Diligence: Perform due diligence on the insurer's financial health, management practices, and regulatory compliance.

Valuation and Pricing:

Accurate Liability Valuation: Conduct a detailed actuarial valuation of the scheme's liabilities to determine the appropriate transfer value.

Pricing of Annuities: Work with the insurer to agree on the pricing of annuities or other financial products used to guarantee the benefits.

Nature of benefits : ensure that the benefits under annuity product match the terms of benefits committed to the member. Else there could arise a requirement of purchase fresh annuities later on.

Taxation:

Manage and seek advice regarding taxes on liquidating the scheme assets.

Risk Management:

Risk Transfer: Assess the extent to which investment, longevity, and other risks are transferred to the insurer.

Contract Provisions: Ensure the insurance contract includes provisions for managing and mitigating risks, such as escalation clauses for inflation adjustments.

Member Communication:

Clear Explanation: Clearly explain the transfer process and its implications to members, ensuring they understand how their benefits will be secured.

Feedback Mechanism: Establish a mechanism for members to provide feedback or raise concerns about the transfer.

Regulatory Approval:

Regulatory Compliance: Ensure the transfer complies with all regulatory requirements and obtain necessary approvals from relevant authorities.

Reporting and Documentation: Maintain accurate records and fulfill all reporting obligations related to the transfer.

Post-Transfer Monitoring:

Ongoing Oversight: Implement ongoing monitoring of the insurer's financial health and performance to ensure the continued security of the benefits.

Contingency Planning: Develop contingency plans in case the insurer encounters financial difficulties in the future.

[10]

[27 Marks]
